

Geometry: Unit 4, Lesson 5

1. A sequence of transformations on point $(-8, 3)$ are described.

$$(x, y) \rightarrow (-y, -x)$$

$$(x, y) \rightarrow (x + 2, y - 1)$$

$$(x, y) \rightarrow (-y, x)$$

What is the x -value of the point after this sequence of transformations?

2. An object on a plane is transformed using a sequence of transformations by first having both its x and y -coordinates become their opposites, $(x, y) \rightarrow (-x, -y)$, and then having its x -coordinate become its opposite, $(x, y) \rightarrow (-x, y)$. What is a written description that represents the transformations?

3. Triangles ABC , $A'B'C'$ and $A''B''C''$ are shown on the coordinate grid. What sequence of algebraic transformations maps triangle ABC onto triangle $A'B'C'$ and then onto triangle $A''B''C''$?

